

Unitree Week 29

Presentation by Mario Ortiz

This Week

- Preparing Labs for Embodied AI

Lab 1: IsaacLab/IsaacSim Setup + R1 Import

In Lab 1, we will focus on these things:

- IsaacLab Setup (download on github)
- IsaacSim Setup (Downloading IsaacSim, bridging IsaacSim to IsaacLab)
- Providing information about Unitree G1 humanoid locomotion directories (provided by isaac lab)
 - `rough_env_cfg.py`: main locomotion task on rough terrain environment
 - `flat_env_cfg.py`: inherits `rough_env_cfg.py` and removes terrain commands + unnecessary rewards/penalties used for rough terrains.
 - `__init__.py`: makes the play/train commands
 - `rsl_rl_ppo_cfg.py`: parameters for training (hidden dimensions, # of iterations, epochs, etc.)
- Show some examples with `train.py` and `play.py` of rough and flat env codes.
- Finally, show setup of the Unitree R1 onto IsaacSim (obtaining the usd file)
 - Download URDF file
 - Export to USD file on IsaacSim

Lab 2: R1 rough_env_cfg.py

Lab 2 Shows comparing R1 code with G1 code. It will show that the two robots although look similar have different properties and therefore must be coded differently.

- Will make similar directory to G1 for the R1.
- Will then show the changes in the R1 code with the G1.
 - Example: The R1 has a different term for the G1's pelvis joint.

Lab 3: R1 flat_env_cfg.py and stand_env_cfg.py

Lab 3 Will similarly show the Unitree R1 code for the flat_env_cfg.py and the differences with that of the G1. In addition, we will show we can manipulate the rough_env_cfg.py by making locomotion have different properties (stand_env_cfg.py shows we can mitigate locomotion properties to teach the robot to simply stand).

- Will show the differences in G1 flat_env_cfg.py with R1 flat_env_cfg.py
- Will then show how stand_env_cfg.py works and why it is useful for training the R1 with other methods (imitation learning).

Lab 4: Stair_env_cfg.py

This lab will show we can train the Unitree to do very specific tasks. In this code, we train the G1 to walk up the stairs. This will also showcase the challenges of developing locomotion tasks and how many factors must be accounted for.

- First we will show how we manipulate the terrain using the IsaacLab's assets, and build a staircase with it.
 - The staircase increases in height as we increase in step.
- Then we will show how specific penalties/rewards need to be changed so the R1 can walk through the obstacle course.
 - Ex. a penalty was made punishes the robot for moving to the left/right of the staircase, ensuring it knows to walk straight only.

